

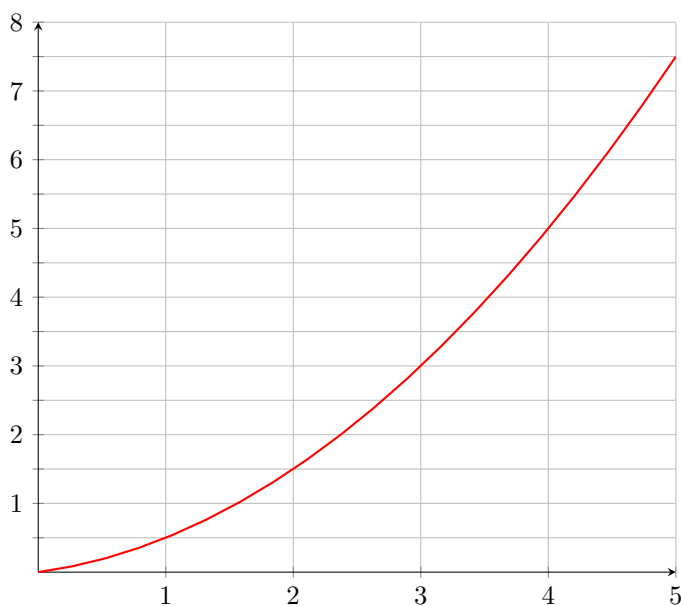
For the week of September 01-05

- 1) A snail crawls up and down a stick. Its velocity $v(t)$, measured in *cm* per minute, is given in the table below.

t	0	5	10	15	20	25	30	35	40	45	50	55	60
$v(t)$	16	-4	-2	9	22	23	11	0	-12	-12	-1	6	17

Use the information in the table to write a Riemann sum that approximates the

- a) total distance travelled by the snail in the first 60 minutes. Your Riemann sum should use 4 rectangles and left endpoints.
- b) net distance travelled by the snail in the first 60 minutes. Your Riemann sum should use 3 rectangles and right endpoints.
- 2) A car is standing at a traffic light. When the light turns green, the car's velocity increases at a constant rate for 12 seconds until it reaches a speed of 60km/h. It travels at a constant velocity for another 10 seconds. As the car approaches the next traffic light, it brakes for 3 seconds and its velocity decreases to 0 at a constant rate. Find the distance between the two traffic lights.
- 3) Two bicyclists begin travelling in the same direction from the same point at time $t = 0$ seconds. Bicyclist A travels with a velocity of $(t + 1)$ m/s. Bicyclist B begins travelling with a velocity of $(2t)$ m/s for 5 seconds and then travels with a velocity of 10m/s. Find all times, in seconds, when the bicyclists are an equal distance from the starting point. How far did each of them travel at that time?
- 4) A tank of water leaks its contents at a rate of $f(t) = (\sin(2t) + 1)$ l/h. The total amount of water leaked on the time interval $[a, b]$ is the area between the graph of $f(t)$ and the t axis on the interval $[a, b]$. Approximate the total amount of water that leaked from the tank in the first 3 hours using a Riemann sum with 6 intervals and left endpoints.
- 5) A graph of a function is given below. Write a Riemann sum that approximates the area under the graph on the interval $[0, 5]$ using 5 rectangles and right endpoints. Sketch the rectangles you are using over the function. Is the number you get an over or under approximation?



- 6) For each function $f(x)$ given below, write a Riemann sum using n rectangles and left endpoints that approximates the net area between the function and the x axis on the given integral.

a) $f(x) = x^3 + 2x^2$ on $[-2, 2]$.

b) $f(x) = \sqrt{2^x}$ on $[0, 10]$.

c) $f(x) = \cos 2x$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

d) $f(x) = \frac{1}{x^2} - 2$ on $[1, 5]$.

- 7) Write each integral below as the limit of a Riemann sum using right endpoints.

a) $\int_2^4 (1 + x^2)^2 dx$.

b) $\int_1^e 2 \ln(3x) dx$.

c) $\int_0^{100} e^{-x^2} dx$.

d) $\int_{-\pi}^{\pi} \sin(x+1) dx$.

8) For each limit of a Riemann sum below, write an integral that is equal to it.

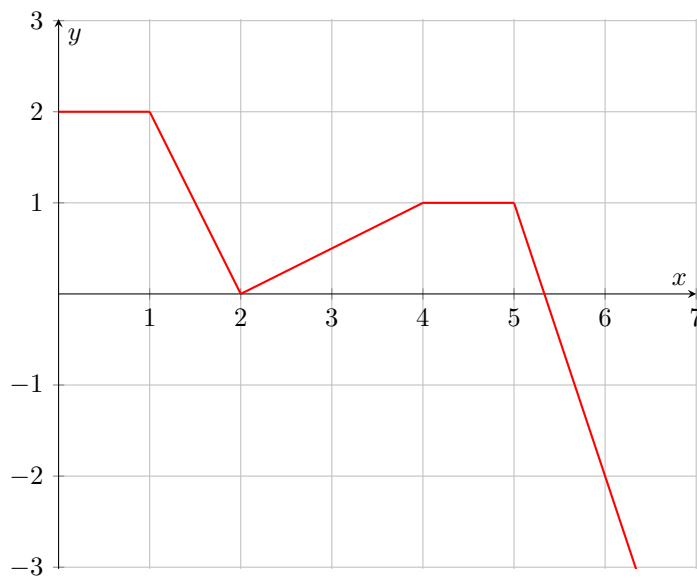
a) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{5}{n} \left(7 - \left(\frac{5i}{n} \right)^2 \right)$.

b) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{5}{n} \left(7 - \frac{5i}{n} \right)^2$.

c) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{5}{n} \left(7 - \left(\frac{5i}{n} \right)^2 \right)^2$.

d) $\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} \frac{1}{n} \left(7 - \left(\frac{5i}{n} \right)^2 \right)$.

9) The graph of the function $f(x)$ is given below. Evaluate $\int_1^6 f(x) dx$.



10) You are told the following information about a function:

- $f(0) = 3$,

- A left hand Riemann sum for $f(x)$ on the interval $[0, 8]$ using 4 subintervals gives a value of 30,
- A right hand Riemann sum for $f(x)$ on the interval $[0, 8]$ using 4 subintervals gives a value of 48.

Use the above information to evaluate the following two quantities.

(a) $\sum_{i=1}^4 f(2i)$.

(b) $f(8)$.

- 11) Suppose that f is an odd function, g is an even function, and that both of them are continuous. Let $\int_{-1}^0 f(x) + g(x) dx = 1$, $\int_{-7}^0 f(x) + g(x) dx = 2$, $\int_0^7 f(x) dx = 3$, and $\int_1^7 g(x) dx = 4$. Evaluate the following:

a) $\int_0^1 f(x) + g(x) dx$.

b) $\int_{-1}^1 2g(x) dx$.

c) $\int_7^{-1} f(x) dx$.

d) $\int_0^7 2f(x) - 3g(x) dx$.

The following problems are for practice only and will not appear on quizzes.

- I) Textbook section 5.1, problems # 29, 30, 36, 37, 39, 40, 41, 42, 43, 45, 49.
- II) Textbook section 5.2, problems # 9, 13, 20, 25, 35, 37, 38, 39-42, 45, 62-67, 72-75, 80.
- III) Textbook section 5.4, problems # 1-6, 27, 29, 30, 31, 32, 33-41.
- IV) Suppose that f is a continuous function and that $\int_a^b f^2(x) dx = 0$. What functions could f be?